

# RAHUL RAJ R

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## OBJECTIVE

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Final-year Electrical Engineering undergraduate with hands-on experience in embedded sensing, optical/EM simulations, and signal acquisition systems. Skilled in MATLAB/Simulink, embedded hardware, and computational modeling. Seeking a research internship in optical systems, computational EM/optics, or embedded hardware for sensing and communication.

## EDUCATION

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|------------------------------------------------------------------------|-------------|
| <b>Bachelor of Technology in Electrical Engineering</b> 8 CGPA         | 2022 - 2026 |
| Vellore Institute of Technology, Chennai Campus, Tamil Nadu            |             |
| <b>Maharishi International Residential School, CBSE, Class XII</b> 80% | 2020 - 2022 |
| <b>Dr.G.S.K memorial school, CBSE, Class X</b> 93%                     | 2013 - 2020 |

## SKILLS

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|-----------------------|---------------------------------------------------------------------|
| <b>Programming</b>    | Python, C, Java, Rstudio, MATLAB                                    |
| <b>Circuit Design</b> | LTspice, PSpice, EasyEDA, Keil $\mu$ Vision, Qflow, PSCAD, Simulink |
| <b>Language</b>       | English, Tamil, German(A1)                                          |

## PROJECTS

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### Axial Flux Permanent Magnet Motor for Robotic Application (Hardware)

Designed and simulated a Halbach-array axial-flux motor (ANSYS Maxwell) with 24-slot/12-pole geometry.

- 3D-printed prototype achieved 1.91 Nm @ 500 rpm, validating high torque density for compact actuation.

### QPSK Transceiver Simulation (Simulink)

Modeled a complete QPSK transmitter–receiver chain and analyzed BER/SNR and constellation distortion.

- Evaluated channel performance relevant to coherent optical communication systems.

### Smart Hydration Tracker using ESP32 (IOT)

Built an ESP32-C3 based miniature weighing coaster with FSR sensing, Wi-Fi telemetry and cloud/mobile logging.

- Added an AI-based hydration predictor using BMI, weather and humidity; designed custom CAD enclosure.

### Self-Balancing Robot using ESP32 (Control & Sensor Fusion)

Implemented IMU-based attitude estimation and PID control on ESP32; designed PCB and sensor–actuator interfaces.

- Tuned closed-loop control for stable balance under varying loads.

### Access Management System (IR, Camera & RFID)

Developed an embedded multi-sensor access-control system integrating camera, IR and RFID modules.

- Implemented low-latency sensor-fusion logic with event logging and basic analytics.

### Real-time Air-Quality Monitoring with AI Forecast (IOT)

Built ESP32-based AQ monitor using MQ-135/DHT11 with wireless data logging and dashboards.

- Implemented ML-based forecasting and anomaly detection from historical sensor data.

### Physical Design of SPI Master Core (Qflow)

Designed RTL for an SPI master and completed synthesis, floorplanning and routing using Qflow/Yosys.

- Performed timing checks and gained exposure to DRC/LVS physical verification.

### Pumped-Storage Hydropower Simulation (MATLAB/Simulink)

Modeled pumped-storage operation and control in MATLAB and Simulink separately to verify results.

- Analyzed efficiency and transient response and compared it with hydro electric power system.

## EXPERIENCES

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### Team Lead – Industrial Energy Audits

Dec 2023 – Mar 2024

*India 4.0 Technologies*

*On-Site & Remote*

- Led multi-person technical teams for industrial energy and power-quality audits across manufacturing sites.
- Personally acquired and processed measurement datasets from the Fluke 437-II Power Quality Analyzer, building experience in precision data capture and time-series analysis.
- Prepared detailed audit reports with corrective recommendations and KPI tracking to improve electrical efficiency.
- Contributed to documentation and proposals for a **government-funded tender** that secured national-level audit contracts for power stations and railways.

### Power Quality Auditing & Standards Analyst

Nov 2023 – Dec 2023

*ICD Power Solution Pvt. Ltd.*

*On-Site*

- Conducted field-level measurements and interpreted results against IEEE and GRI standards for compliance assessment.
- Analysed site recordings, identified anomalies and supported corrective-action planning based on measured parameters.
- Compiled technical reports and presented findings to stakeholders, reinforcing clear data-driven communication.